

MGT 3213 – Management Information Systems

S # 6 - Information Systems Development

Dr. A. Rukshan

Senior Lecturer in IT

Dept of Management and Entrepreneurship

Faculty of Business Studies

University of Vavuniya

<https://www.arukshan.com/>

Information Systems Development (ISD)

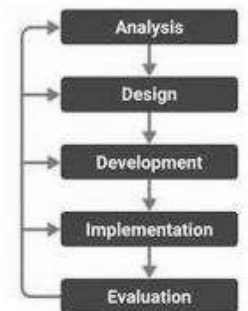
- Information Systems Development (ISD) is the comprehensive process of:
 - planning,
 - creating,
 - implementing, and
 - maintaining information systems (like software, databases, networks)
- to support an organization's strategic goals, involving stages like analysis, design, development, testing, and deployment, using various methodologies, for example Agile, Waterfall, to manage technical, human, and business aspects.
- It focuses on building solutions, from custom software to integrated platforms like ERP/CRM, ensuring they align with business needs and deliver value.

Agenda

- Stages in ISD
- Methodologies for ISD
 - Waterfall Methodology
 - Agile
 - V-Model
 - Spiral Model

Stages in ISD

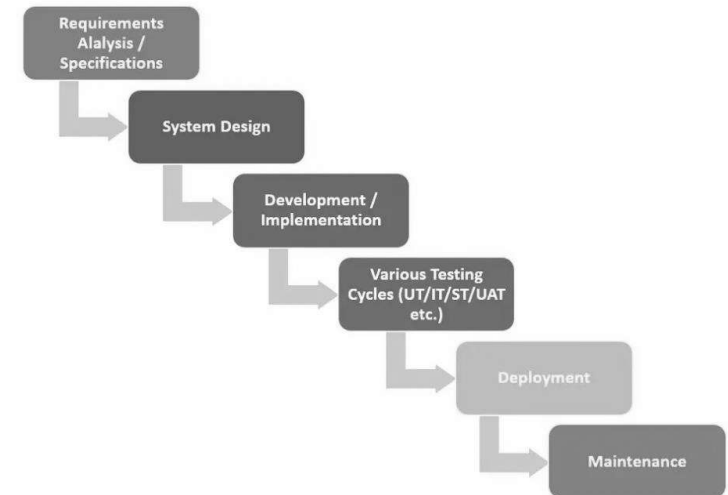
- **Planning & Analysis:**
 - Defining needs, objectives, feasibility, and gathering requirements.
- **Design:**
 - Creating blueprints for the system, including architecture, user interfaces, and data structures.
- **Development/Implementation:**
 - Writing code, configuring software, and building the system.
- **Testing:**
 - Verifying the system works correctly and meets requirements.
- **Deployment/Maintenance:**
 - Launching the system and providing ongoing support and updates.



Methodologies for ISD

- **Waterfall:**
 - Sequential, linear approach.
- **Agile:**
 - Iterative and incremental, focusing on flexibility and collaboration.
- **V-Model:**
 - Extends Waterfall with testing phases aligned with development stages.
- **Spiral Model:**
 - Risk-driven, iterative approach.

Waterfall Methodology (for Software Development Project Management)



Waterfall - Requirement Analysis Stage

- **Challenges:**
 - Understanding user needs accurately; avoiding scope creep.
- **Best Practices:**
 - Use of detailed questionnaires, stakeholder interviews, and feedback sessions.
- **Tools & Techniques:**
 - Requirements management tools, flowcharts, and use-case diagrams.

Waterfall - Design Phase

- The design phase has two sub-phases as follows:
 - **Functional design** – (or **Logical Design**) is in the hands of the analysts who use the information from the Requirements phase to design the system
 - **Technical specifications design** – the functional design is transformed the system according to technical specifications or physical design.
- **Challenges:**
 - Balancing user requirements with technical feasibility.
- **Best Practices:**
 - Prototyping, peer reviews, and design validation.
- **Tools & Techniques:**
 - Design modeling tools, wireframing, and architectural diagrams.



Waterfall - Development Phase

- Writing the code for development projects or customization for software package implementation projects.
- The programmers or the Customization team will take control in this phase, taking and understanding the project requirements and specifications to create the code for the application
- **Challenges:**
 - Adhering to specifications; managing technical debt.
- **Best Practices:**
 - Code reviews, continuous integration, and modular development.
- **Tools & Techniques:**
 - Integrated development environments (IDEs), and version control systems.

Waterfall - Testing Phase

- Ensuring that the development has gone through the Verification or testing (quality assurance)
- Various test cycles like Unit Testing, Integration Testing, System Testing, User Acceptance Testing, etc., are carried out to ensure the project meets customer expectations.
- **Challenges:**
 - Identifying and fixing defects and ensuring test coverage.
- **Best Practices:**
 - Automated testing, test-driven development, and regression testing.
- **Tools & Techniques:**
 - Test automation tools and bug-tracking systems.

Waterfall - Maintenance Phase

- Customer interacts with the developed application.
- This is an ongoing phase until the software remains in use.
- This phase takes care of bug fixing or implementing any changes in the requirements and system enhancements, etc.
- In the case of packaged software, this phase also covers version upgrades.
- **Challenges:**
 - Efficiently handling bugs and new feature requests.
- **Best Practices:**
 - Regular updates, user feedback loops, and proactive maintenance.
- **Tools & Techniques:**
 - Ticketing systems and performance monitoring tools.

Pros of Waterfall Project Management

- Simple and Easy to Understand
- Well-Defined Stages
- Emphasis on Documentation
- Easy to Manage
- Predictable Outcomes

Cons of Waterfall Project Management

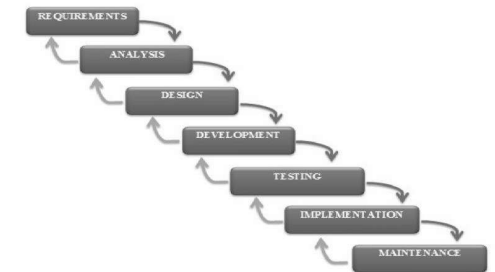
- Inflexibility
- Late Testing and Integration
- Poor Adaptability to Change
- Delayed Delivery
- Risk of Misalignment with User Needs
- Higher Risk and Uncertainty

Adaptations and Variations of the Waterfall Model

- the Waterfall model has faced criticism for its rigidity and lack of flexibility.
- To address these limitations, several adaptations and variations have been developed:

1. Modified Waterfall Model:

- This model incorporates feedback loops, allowing for minor revisions to previous phases.
- This approach acknowledges the need to revisit and revise stages based on new information or changes.



Adaptations and Variations of the Waterfall Model

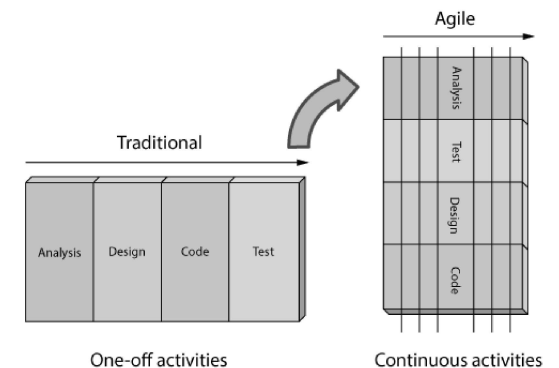
• Sashimi Model (Waterfall with Overlapping Phases):

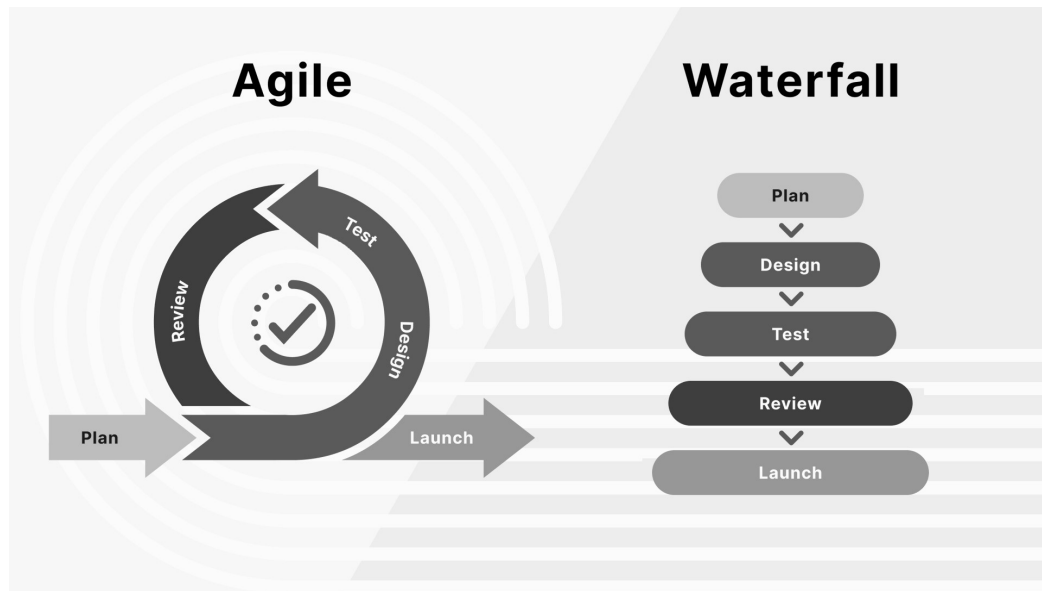
- This methodology allows for slight overlaps between phases. For instance, the design phase can start before the complete finalization of the requirements phase, facilitating a more fluid transition between stages.

Adaptations and Variations of the Waterfall Model

• Agile-Waterfall Hybrid:

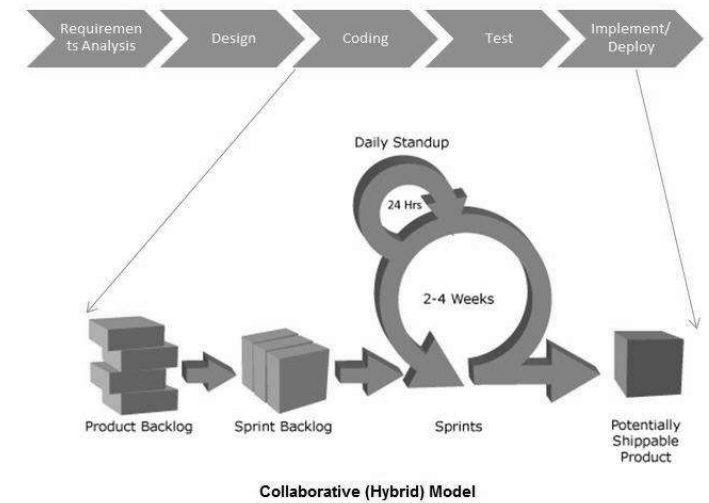
- This model combines the structured approach of Waterfall with the flexibility of Agile methodologies.
- In this model, planning and requirements gathering are conducted in a Waterfall manner.
- However, development and implementation are conducted using Agile practices, allowing for iterative testing and refinement.



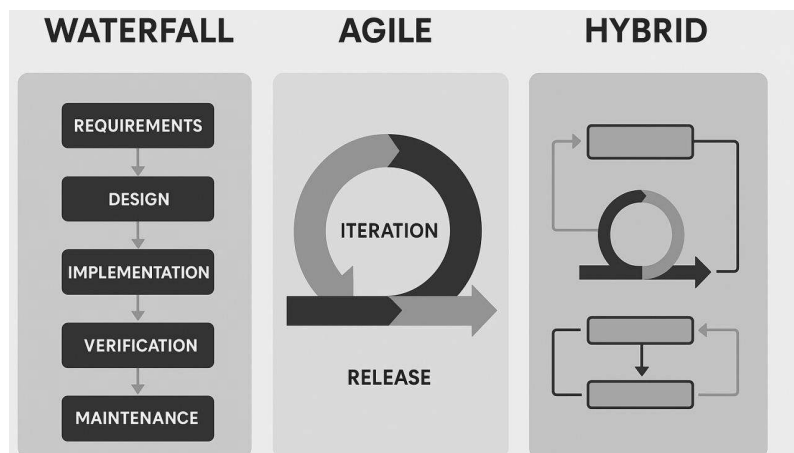


Adaptations and Variations of the Waterfall Model

• Agile-Waterfall Hybrid



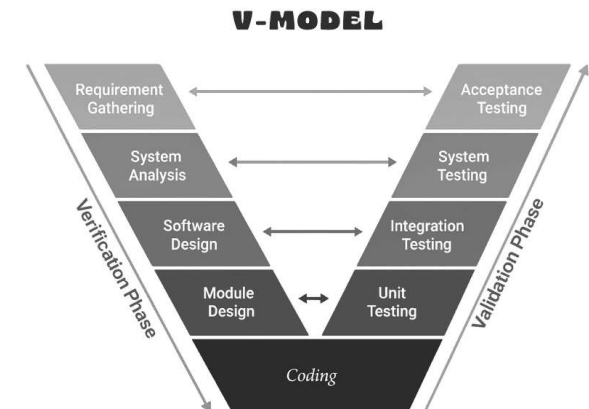
Agile-Waterfall Hybrid



Adaptations and Variations of the Waterfall Model

• V-Model (Validation and Verification Model):

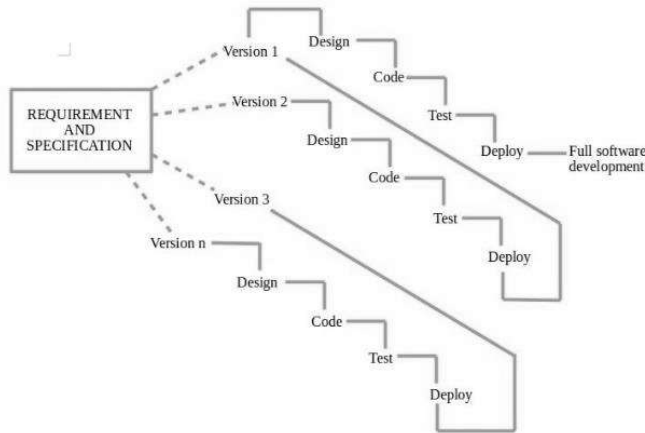
- An extension of the Waterfall model, where each development stage is directly associated with a testing phase.
- This model emphasizes verification and validation parallel to each stage of development, enhancing quality control.



Adaptations and Variations of the Waterfall Model

• Incremental Waterfall Model:

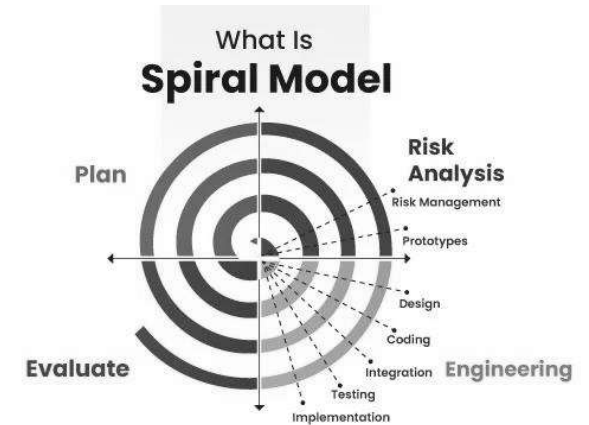
- This model divides the project into smaller segments, each following the Waterfall phases.
- This allows portions of the project to be delivered incrementally, facilitating earlier testing and feedback.



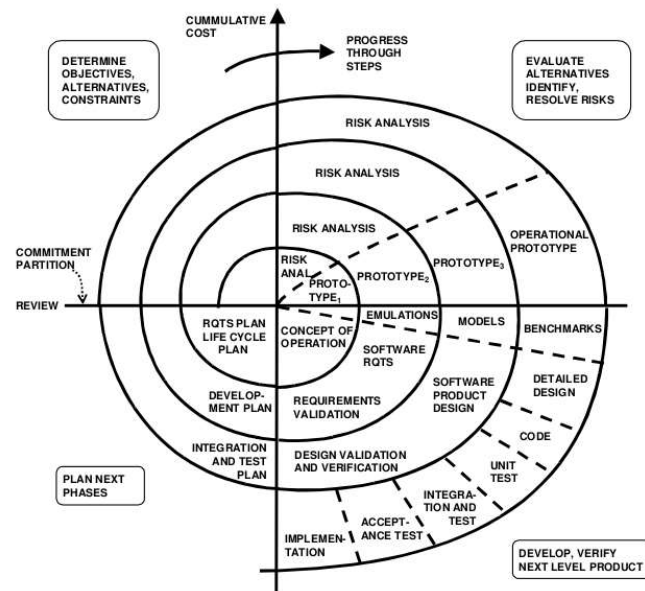
Spiral Model for IS Development

Four phases of Spiral model for software development are:

1. Planning
2. Risk Analysis
3. Engineering
4. Evaluation



Spiral Model



Advantages Of Spiral Model

- Best Model To Handle Risky Projects
- Good Choice For Large Projects
- Flexible With Requirements
- Better Customer Satisfaction
- Iterative and incremental approach
- Open to receiving any feedback
- Better quality

Disadvantages Of Spiral Method

- Complexity
- Expensive
- Too much dependency on experts
- Poor time management
- Follows multiple iterations
- Time-consuming
- Require intensive investment

References

- <https://www.easyredmine.com/news/waterfall-vs-agile-what-is-the-best-approach-for-a-software-development-project>
- <https://www.linkedin.com/pulse/waterfall-vs-agile-hybrid-choosing-right-project-management-dlyvf/>
- <https://birdviewpsa.com/blog/stop-the-battle-how-forward-thinking-organizations-benefit-from-agile-waterfall-hybrid/>
- <https://www.ejable.com/tech-corner/methodologies/waterfall-methodology/>
- <https://www.bdtask.com/blog/spiral-model-for-software-development>
- <https://xbsoftware.com/blog/software-development-life-cycle-spiral-model/>

When to Choose Each Software Development Methodology

Methodology	Best Used When	Project Characteristics	Advantages	Not Suitable When
Waterfall Model	Requirements are clearly defined and stable from the beginning	• Small to medium projects • Low uncertainty • Well-documented processes • Regulated environments	• Simple and structured • Easy to manage • Clear milestones	• Requirements are likely to change • Need fast customer feedback • Innovation-driven projects
Agile Methodology	Requirements are uncertain or evolving	• Startups & digital products • Customer-centric solutions • Fast-changing markets • Innovation and experimentation	• High flexibility • Faster time-to-market • Continuous customer feedback	• Highly regulated systems • Large teams needing heavy documentation
V-Model	High emphasis on testing and validation	• Safety-critical systems • Healthcare, banking, aviation • Compliance-heavy projects	• Early testing planning • High quality & reliability • Clear verification & validation	• Requirements change frequently • Exploratory or innovative projects
Spiral Model	Project involves high risk and complexity	• Large-scale systems • New or unproven technologies • Risk-intensive projects	• Strong risk management • Combines prototyping and planning • Suitable for complex systems	• Small or low-risk projects • Limited budget or expertise

Thank you